

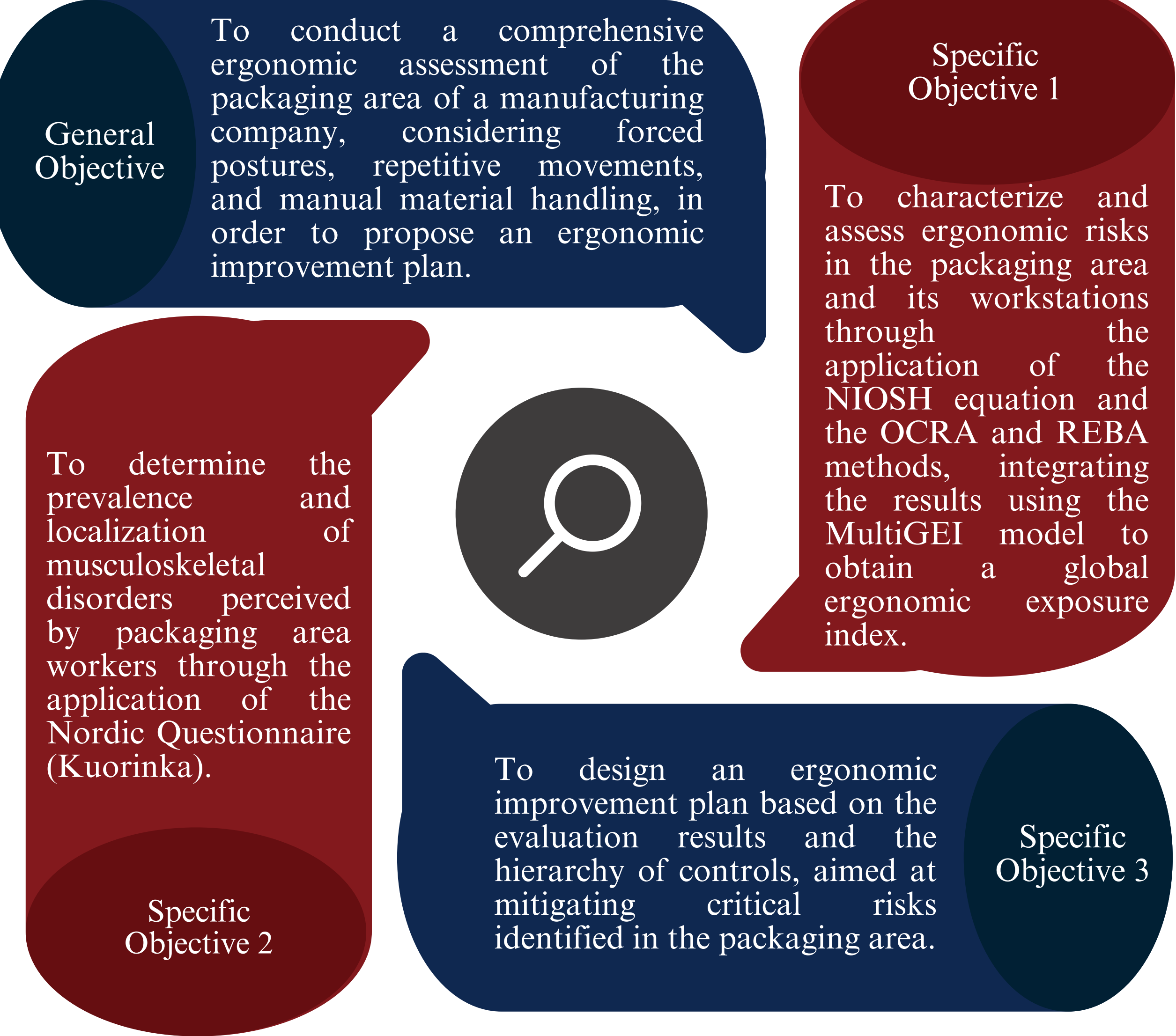
Comprehensive ergonomic assessment of the packaging area of a non-metallic mineral production company, considering the risks associated with repetitive movements, manual material handling, and forced postures.

- Jhon Walter Farez Plasencia
- **Advisor:** Eng. Paulina R. Espinoza Hernández, MSc

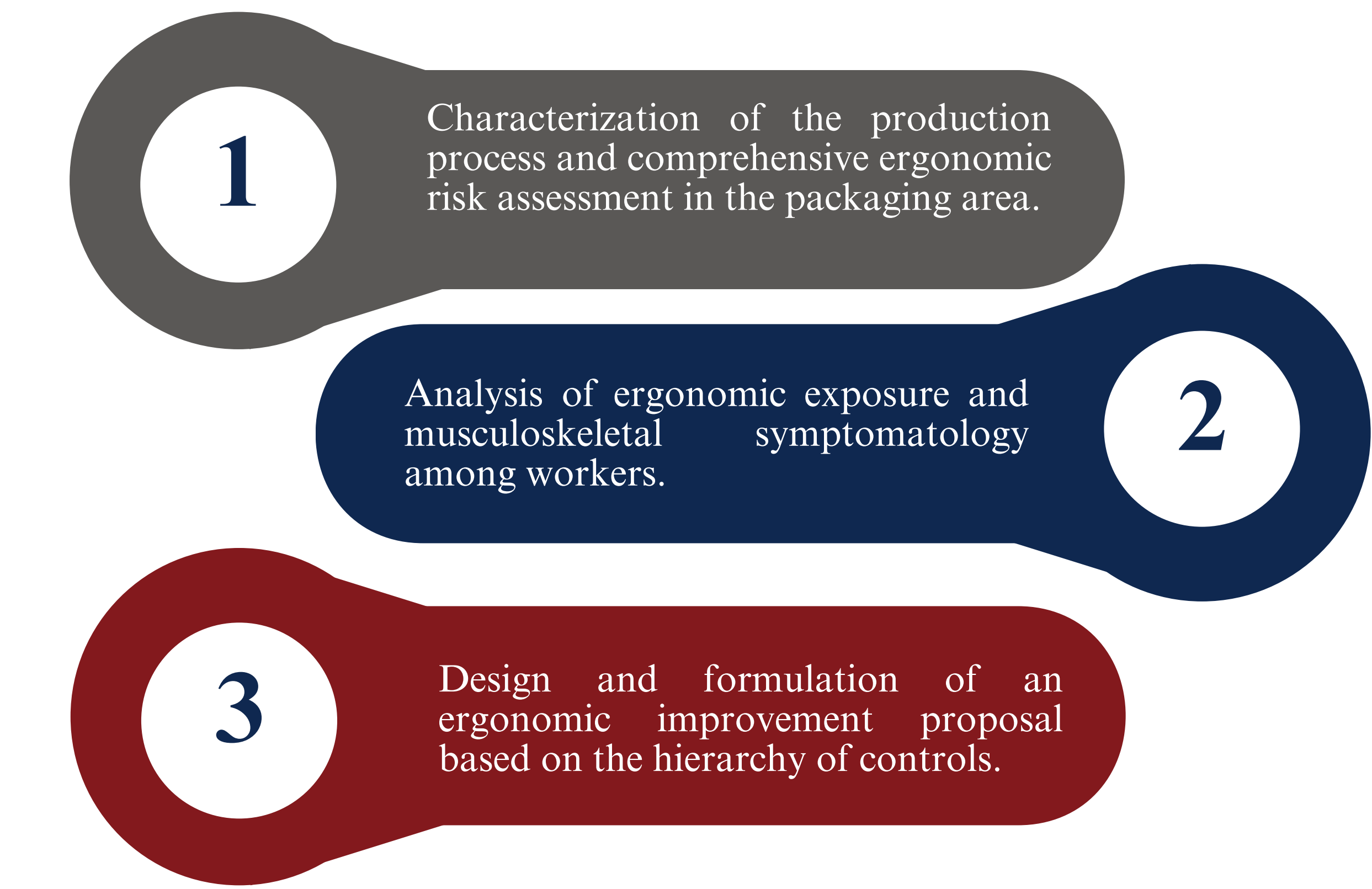
Abstract

Ergonomic risks in the packaging area were assessed using NIOSH, OCRA, and REBA, supported by the MultiGEI model and the Nordic Questionnaire. High risks related to manual load handling and repetitive movements were identified, leading to the proposal of engineering-based improvements aligned with ISO 45001:2018.

Objectives



Methodology



Curricular Integration

Process Engineering and Ergonomics: Ergonomic assessment methods (NIOSH, OCRA, REBA). **Occupational Safety and Health:** Management and evaluation of ergonomic risks.

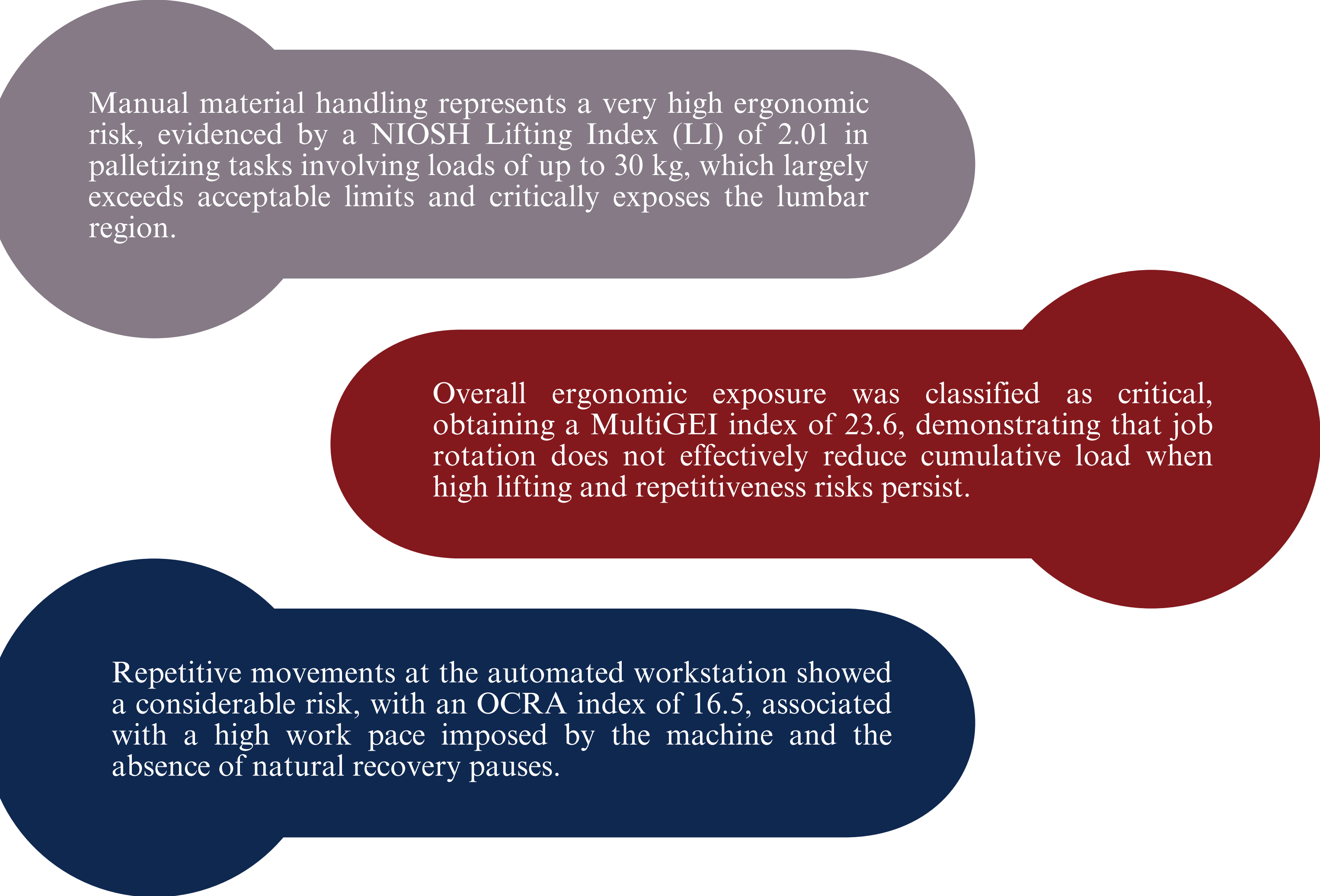
Applied Engineering

Executive Decree No. 255: Occupational Safety and Health. **Technical Standard (Annex 3):** Forced postures and manual handling. **ISO 45001:2018:** OHS management.

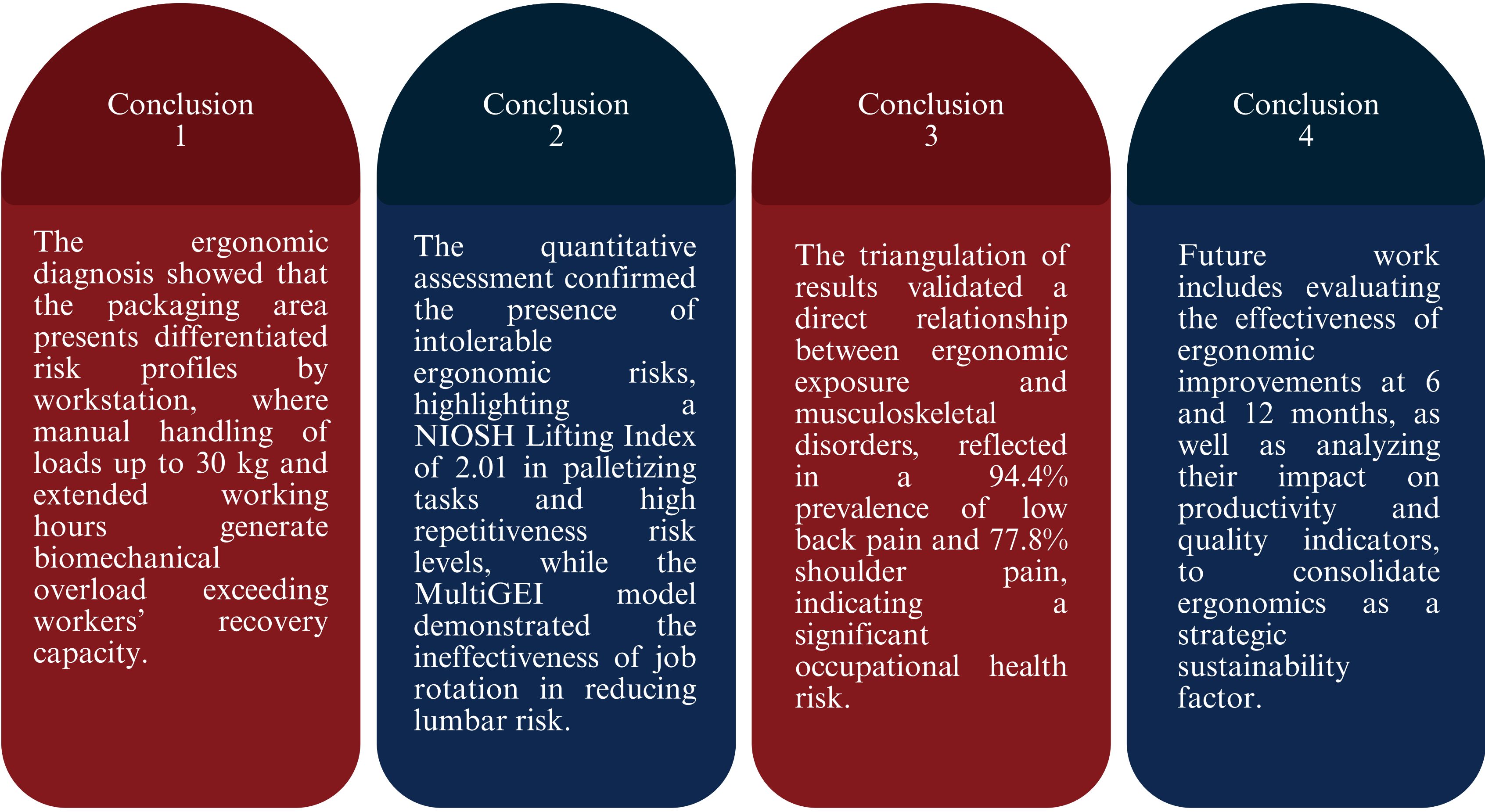
Keywords

- Ergonomics
- Risk Assessment
- Repetitive Movements
- Musculoskeletal Disorders
- Manual Material Handling
- Forced Postures

Results



Conclusions



Contact Information

•Farez Plasencia, J. W.
Email: jhon.farezp1@ucuenca.edu.ec
Mobile number: 0967264961



References

(Colombini et al., 2021), (Waters et al., 1993), (Kuorinka et al., 1987), (INSST, 2021), (Presidencia de la República del Ecuador, 2024), (Rojas Silva et al., 2024), (Trstenjak et al., 2025)